

IFYFM004 2024/25

NCUK
THE UNIVERSITY CONSORTIUM

FURTHER MATHEMATICS

INTERNATIONAL FOUNDATION YEAR (IFY)



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SYLLABUS OVERVIEW

Module Code	IFYFM004
Module Name	Further Mathematics
Programme Name	International Foundation Year
Percentage breakdown of Coursework	0%
Percentage breakdown of Exam/Test	100%
Delivery period	The syllabus will usually be delivered over two 15-week semesters
Recommended minimum teaching hours	120 (4 hours per week for 30 weeks)
Recommended minimum independent study hours	120 (4 hours per week for 30 weeks)

AIMS

To develop students' key knowledge, understanding, skills and application of further mathematics appropriate for entry to a degree course at an NCUK University.

General Aims

- ⇒ To develop abilities to think logically, to recognise incorrect reasoning and to express ideas clearly.
- ⇒ To develop an enthusiasm for the subject and the skills required to apply the knowledge to both the further study and application of mathematics.
- ⇒ To develop in students an understanding of how theory and application work together.
- ⇒ To develop students' skills in modelling and the interpretation of results.
- ⇒ To develop the necessary English mathematics vocabulary and terminology to use their mathematics knowledge effectively in an NCUK University.
- ⇒ To encourage students towards a level of independence in both the planning and organisation of their studies.
- ⇒ To assist the development of competence and confidence of the students as learners, taking responsibility for their own learning through directed reading and study.

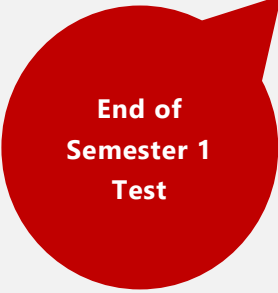
Specific Aims

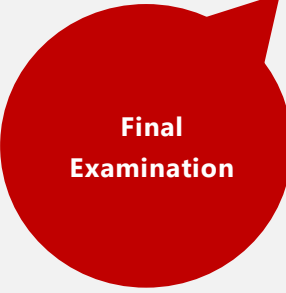
- ⇒ To develop further skills in algebra and link these with curve sketching where possible.
- ⇒ To demonstrate more advanced skills in complex numbers, matrices, vectors, hyperbolic functions and geometry (conics).
- ⇒ To develop differentiation and integration techniques and to apply these to differential equations.
- ⇒ To apply techniques from pure mathematics to the mechanics topics.
- ⇒ To apply a Mathematical proof.
- ⇒ To use numerical methods with the aid of calculators.

TOPICS OF STUDY

- ⇒ Complex Numbers
- ⇒ Matrices
- ⇒ Further Curve Sketching and Inequalities
- ⇒ Series
- ⇒ Roots and Coefficients of Polynomials
- ⇒ Mechanics
- ⇒ Hyperbolic Functions
- ⇒ Parametric Coordinates
- ⇒ Conic Sections
- ⇒ Maclaurin and Taylor Series
- ⇒ Further Mechanics
- ⇒ Further Complex Number Functions
- ⇒ Further Differentiation and Integration
- ⇒ Vectors
- ⇒ Calculus and Vectors
- ⇒ Differential Equations
- ⇒ Proof by Induction

ASSESSMENT

	What is Assessed?	Topics A-J
	Duration/Word Count	2 hours 10 minutes
	Total Marks	100
	Rubric	⇒ Approximately 13 questions of varying length. Candidates attempt all questions.
	Contribution to Overall Grade	30%

	What is Assessed?	Topics A-Q
	Duration/Word Count	2 hours 40 minutes
	Total Marks	125
	Rubric	⇒ Approximately 15 questions of varying length. Candidates attempt all questions.
	Contribution to Overall Grade	70%

ASSESSMENT OBJECTIVES

	Description	End of Semester 1 Test (30%)	Exam (70%)	Overall
AO1	Knowledge & Understanding: Knowledge of mathematical techniques.	50%	50%	50%
AO2	Application and Communication: Recognising the appropriate procedure and applying combinations of skills and techniques.	50%	50%	50%

GENERAL LEARNING OUTCOMES

On successful completion of this module, a student will be able to:

Knowledge and understanding	⇒ Recognise, recall and apply specific mathematical facts, principles and techniques; and employ them in relevant and correct settings. ⇒ Use and interpret their own visual aids to enhance understanding.
Intellectual skills	⇒ Break down problems into smaller sub-problems and devise strategies for solving new problems. ⇒ Select and apply relevant techniques to problem solving in a systematic way. ⇒ Relate new problems to already solved problems. ⇒ Employ critical analysis of solutions. ⇒ Adapt, diversify and re-evaluate methods of solution.
Practical skills	⇒ Carry out appropriate calculations using a formula booklet, and/or a calculator. ⇒ Use technology to enhance understanding of graphs and loci.
Transferable skills	⇒ Study independently and make personal notes for problem-solving and revision purposes. ⇒ Source and retrieve information from a variety of locations such as textbooks, websites etc. ⇒ Present information in a clear and logical way. ⇒ Discuss, interpret and criticise results.

TEACHING, LEARNING AND ASSESSMENT GUIDANCE

Teachers should use a range of different learning and delivery styles in order to give students experience of the types of approach they will encounter in an NCUK University e.g. lectures, tutorials, seminars, laboratory practicals etc. Appendix A (Exemplar Teaching Plan) suggests a delivery format designed to facilitate teaching and learning in each topic. This is provided as a guide and teachers are at liberty to diverge from this plan. The Study Centre may also choose to timetable more than the minimum number of teaching hours (4 hours per week) to meet the needs and abilities of students.

Teachers should take every opportunity to use case study and data response or practice material to develop the ability of students to apply their knowledge and develop their skills of handling and evaluating information. Tutorials should involve both group and individual activities, with a strong emphasis on applying knowledge from lectures and independent study to problem-solving. It is important for all students to have opportunities to speak in English during each tutorial. Suggested activities include students being encouraged to explain in English their answers in pairs or small groups, students providing answers to the whole class whilst standing at the front of the group, student-led presentations, and group activities that require discussion and debate.

Students will have different backgrounds in the subject and it will be necessary to provide opportunities for directed independent study, so as to allow each student to develop at their own pace to reach the required level for the assessments. Activities (homework) for independent study should be set weekly. It is advisable to also consult with EAP teachers when planning homework as there may be subject tasks which can support the development of academic English (and vice-versa).

DEVELOPING INDEPENDENT LEARNING

It is important that the learners are given the guidance and support to develop from pupils within a teacher-led learning environment, to students who feel confident taking ownership of their learning through independent study. As students may have an educational background where concepts of teaching and learning are different to those prevalent in NCUK Universities, aspects of independent learning should be introduced gradually throughout the programme.

Below are suggested examples of ideas which could be used to encourage independent learning:

Acknowledging Failure and Success

A fear of failure or making mistakes can be a significant obstacle which prevents learners from being independent. It is therefore important to acknowledge the benefits of failure with regard to learning early in the module delivery. This could be explored in more depth through practical examples and calculations in which failure leads to success. Teachers should ensure that students are also praised, particularly through formative feedback, and encourage learners to demonstrate development from their past mistakes.

Classroom Reversal

Where possible, teachers might entrust the teaching, practice and revision of sub-topics or items from the syllabus to students themselves. Sharing the responsibility of teaching with students can be a more motivating way to encourage them to take their first steps into independent study than being asked to

study independently for purely personal reasons. It also provides a strong reason why independent learning should be carried out (in this case, to support their peers) and a visible outcome of their independent work (the performance of their peers).

A demonstration of possible student-led tasks and activities can be a useful way to introduce this concept within the classroom. A clear schedule for student-led sessions can also help to make it routine. Sessions could be led by groups initially before asking students to lead them individually when they have grown in confidence.

PAST ASSESSMENTS

Many learners can find it reassuring to revisit past assessments to support their preparation for formative and summative assessments. A selection of past Coursework, End of Semester 1 Test and Final Examination papers are available for this purpose via the NCUK Teachers' Hub. It is important that students are guided to sections and questions which will support the meeting of current learning outcomes or that past assessments are appropriately adapted to ensure their fit to the current syllabus.

While classroom time can be allocated to the review of and use of past papers, students should also have the opportunity to make use of these for independent study. Teachers should work with colleagues to decide which papers may be kept aside for use in formative assessment or class time and which to release to students for independent study.

There are a number of ways past assessments can be used to support learning in addition to recalling knowledge. These include:

- ⇒ An exploration of incorrect or incomplete answers which could highlight gaps in understanding and help to inform future teaching and learning
- ⇒ Identifying whether the correct structure is being used to answer extended answers
- ⇒ Practicing answers to certain questions within particular time constraints
- ⇒ Distinguishing different command words, such as 'explain', 'describe' and 'evaluate'
- ⇒ Allowing students to notice weaknesses in their own knowledge or answers by allowing them to self-mark with the assessment criteria
- ⇒ Allowing students to establish a greater familiarity with the assessment criteria and what they are expected to do to achieve a certain grade

SUPPORTING ENGLISH LANGUAGE DEVELOPMENT

It is important for the development of academic literacies that the delivery of the subject material is integrated with the English for Academic Purposes (EAP) module (or EAP for Proficient Users or Research and Communication Skills modules where appropriate). Regular communication between the subject module teacher(s) and the EAP, EAPPU or RCS teacher(s) will provide a basis on which to support and guide students. Students will benefit from collaborative activities, where the subject module and EAP/EAPPU/RCS teachers jointly deliver classes in relation to activities such as essay or report writing style and using academic sources.

There are also a number of simple tasks and approaches which can be built into this module which can support language development. These include:

Introducing New Vocabulary

When students are introduced to new vocabulary in class, ensure that these words are not introduced in isolation, but in a chunk or phrase within a wider understood context. As an example, if students were given the following extract from the Economist (2021), they may be unfamiliar with the word 'ailments':

The cholera pandemic of the early 1830s hit France hard. It wiped out nearly 3% of Parisians in a month, and hospitals were overwhelmed by patients whose ailments doctors could not explain.

If this word was then to be explored and recorded, it would be better to look at this chunk:

...hospitals were overwhelmed by patients whose **ailments** doctors could not explain.

By giving students a clear context, it is easier to initially discern meaning while also making it more memorable for revision. By keeping the word within a larger chunk, there are also a number of grammatical and lexical relationships which can be established by learners:

- ⇒ Common collocations; explain ailments, a patient's ailments.
- ⇒ The inclusion of 'ailments' in the semantic field of medicine / health.
- ⇒ The determiner 'whose' implies that 'ailments' is a noun and the 's' that it is a plural, countable noun.
- ⇒ Wider examples of past passive and defining relative clauses.

Although it should not be expected that this level of exploration is conducted within a subject module, highlighting new vocabulary in chunks does give language learners a greater opportunity to notice features of language at their own speed. It also provides a grammatical template which students can build on when writing their own answers.

Highlighting Stressed Syllables

One way to support the development of students spoken fluency and overall rhythm is to draw attention the stressed syllable within key, subject-specific vocabulary. This can be done very simply on the board in a number of ways and you may wish to discuss this with the EAP/RCS teacher(s) at your Study Centre in order to align your preferred method(s). As you may be students' main model for how to pronounce subject-specific vocabulary, it is essential they have a clear record which they can refer to during independent study.

Encouraging Functional & Discursive Language

When completing written and spoken tasks, such as group discussions or the presentation of ideas within class, you may wish to encourage students to use functional or discursive language that will help them to structure and explain their ideas with greater coherence and clarity. For instance, this may include language for agreeing /disagreeing, clarifying what they have said when they are unclear, or how to highlight a particular point. This could be achieved by:

- ⇒ Asking students to take out, look at and use language explored for this purpose within their EAP classes. Collaboration with the EAP teacher(s) to decide when in the module particular language will be covered and for what purpose will greatly support students' language development.

- ⇒ Make functional and discursive language relevant to your subject available in the classroom for students to use. Adding useful language to walls or 'vocabulary mats' on desks can remind students of language available to them while also encouraging independence.

Allowing Learners Time to Speak

Although some students may be reluctant to speak in class, it is also important to remember that less proficient speakers require time to construct utterances and may need some time to prepare themselves. While silence in a classroom can sometimes feel awkward, try to avoid filling this silence with teacher talk time, particularly when you are expecting answers from students. You may want to prepare students to speak by giving them advance warning, for example, by giving the class time to make notes before being asked to speak. Alternatively, you may wish to select students to answer particular questions in advance.



FORMATIVE ASSESSMENT

It is important that students are given the opportunity to engage in and submit formative assessments and receive feedback on this work. Formative assessment should be designed to inform students of their progress and enable them to develop and practice coursework and examination skills.

In order to assist teachers in setting formative assessment, NCUK provides the following for this module:

- ⇒ Exemplar assessments – a full set of exemplar assessments with mark schemes (End of Semester 1 Test, Coursework and Final Examination) are downloadable from the Teachers' Hub. These can be used as mock assessments for students throughout the academic year.
- ⇒ Past exam questions – a set of past examination papers is accessible for teachers via the Teachers' Hub. Teachers can generate their own papers for use as practice exams, pick out specific questions to work through with students in class, or set as homework.

In addition to the above, teachers should set regular homework tasks that allow students to understand their strengths and weaknesses and should provide advice to students on how to improve in areas of weakness.

Beyond developing students' skills, the crucial element of formative assessment is the feedback provided by the teacher. In some instances, it may also be appropriate for students to provide feedback on each other's work in a group setting.



SUMMATIVE ASSESSMENT

The breakdown of summative assessment for this module and the rubrics for each component are given on page 3 of this document. All summative assessments are provided by NCUK.

End of Semester 1 Test

The End of Semester 1 Test will cover topics from Semester 1 only and will take place under examination conditions. Teachers at the Study Centre will be responsible for marking the End of Semester 1 Tests in accordance with the NCUK issued mark scheme.

Examination

The examination will cover topics from both semesters. Teachers at the Study Centre will be responsible for marking the Examination in accordance with the NCUK issued mark scheme.

An NCUK approved calculator is permitted in the examination (refer to NCUK policy '*Calculator Regulations*').

It is the Study Centre's responsibility to provide the following materials for the End of Semester 1 test and Examination:

- ⇒ Graph Paper
- ⇒ Formula Booklet (refer to *Appendix C* of this syllabus)

Supporting Documents

It is essential that coursework and examinations are administered in accordance with NCUK regulations. Please refer to the following sections of the NCUK Academic Handbook for details:

- ⇒ Coursework Administration and Regulations
- ⇒ Centre Marking and Recording Results
- ⇒ Academic Misconduct Policy
- ⇒ Examination Administration

RESOURCES



Core Textbooks

CGP Books (2017). *Mathematics for A level The Textbook*, Coordination Group Publications Ltd (CGP)

ISBN: 9781782947233 (print)

Electronic version available to purchase [here](#)

CGP Books (2018), *AS and A level Further Maths for Edexcel Complete Revision and Practice*, Coordination Group Publications Ltd (CGP)

ISBN: 9781782948698 (print)

Each copy comes with a free online edition.



Further Reading

Courant, R. & Robbins, H. (1996), *What is Mathematics? An Elementary Approach to Ideas and Methods*, 2nd Ed. Oxford University Press

ISBN: 9780195105193 (print)

ISBN: 9780199754878 (eBook)

Riley, K.F., Hobson, M.P. & Bence, S.J. (2006), *Mathematical Methods for Physics and Engineering*, 3rd Ed. Cambridge University Press

ISBN: 9780521679718 (print)

ISBN: 9781139637374 (eBook)

Centre for Innovation in Mathematics Teaching A-level Course Material:

<https://www.cimt.org.uk/projects/mepres/alevel/alevel.htm>



Useful Websites

Free mathematics software:

<http://www.geogebra.org/>

<https://www.desmos.com/>

A variety of resources sorted by topic:

<https://www.mathcentre.ac.uk/>

A level maths resources:

<https://www.stem.org.uk/resources/curated-collections/secondary-and-level-mathematics-0>

A level maths revision resources:

	https://app.senecalearning.com/courses?Price=Free&Subject=Maths&Age+Group=A+Level_ For students and teachers to pose or answer any maths problem at any level: http://math.stackexchange.com
 Teacher Resources	NCUK Teacher's Hub – accessible via the NCUK learning and assessment platform – please contact your Study Centre Academic Manager for access.

MODULE CONTENT

All topics must be covered. Appendix A provides an Exemplar Teaching Plan which indicates the proportion of time to be spent on each topic and activities to support student learning.

The table below details the topics and sub-topics to be covered. A set of specific learning outcomes (SLOs) are given for each topic.

Please note: other than the differentiation and integration of $\sinh x$ and $\cosh x$, any calculus in Semester 1 will be confined to IFY Maths topics G and H.

TOPIC A – COMPLEX NUMBERS		
CONTENT	SLOs	
⇒ Imaginary and complex numbers, Argand diagram, Cartesian form. ⇒ Sum, difference, product and quotient of complex numbers. ⇒ Modulus-argument form [aka polar form or (r, θ) form]. ⇒ Conversion between Cartesian and modulus-argument form. ⇒ Conjugate complex roots of polynomials with real coefficients. ⇒ Loci and regions on an Argand diagram.	A1	Define the terms “imaginary number” and “complex number”.
	A2	Show how a complex number can be represented on an Argand diagram.
	A3	Add and subtract complex numbers.
	A4	Evaluate and simplify powers of complex numbers.
	A5	Multiply and divide two complex numbers.
	A6	Determine the modulus-argument form of a complex number.
	A7	Convert from modulus-argument form to Cartesian form, and vice versa.
	A8	Know that if z_1 is a root of a polynomial with real coefficients then z_1^* is also a root where z_1^* is the complex conjugate of z_1 .
	A9	Loci such as $ z - a = b$, $ z - a = k z - b $ and regions such as $ z - a \leq b$, $ z - a \leq z - b $ Loci such as $\arg(z - a) = \beta$, $\arg\left[\frac{z - a}{z - b}\right] = \beta$ will not be included.

TOPIC B – MATRICES		
CONTENT	SLOs	
⇒ Definition of a matrix. Size of a matrix. Row and column vectors. Addition, subtraction and products of matrices. ⇒ Special matrices: null (Z), identity (I), diagonal and transpose. ⇒ Determinant of 2x2 and 3x3 matrices; inverse of a 2x2 matrix. ⇒ Eigenvalues, eigenvectors and the characteristic equations of 2x2 and 3x3 matrices. Transformations are not included.	B1	Construct a matrix of specified size.
	B2	Recognise special matrices I and Z.
	B3	Explain and demonstrate matrix arithmetic (add/subtract, scalar multiple, matrix-matrix and matrix-vector products).
	B4	Find the transpose of a matrix.
	B5	Find the determinant of a 3x3 and a 2x2 matrix.
	B6	Find the inverse of a 2x2 matrix.
	B7	Find the characteristic polynomial of 2x2 and 3x3 matrices.
	B8	Find eigenvalues and eigenvectors of 2x2 and 3x3 matrices.

TOPIC C – FURTHER CURVE SKETCHING AND INEQUALITIES		
CONTENT	SLOs	
⇒ Sketching graphs of the form $y = \frac{f(x)}{g(x)}$ where $f(x)$, $g(x)$ are linear or quadratic functions. ⇒ Finding intercepts with the co-ordinate axes, asymptotes, behaviour near $x = 0$ and for numerically large x. Bounds on y values (non-calculus method). ⇒ The solution of inequalities by algebraic and graphical methods. ⇒ This section builds on the sketching and inequalities covered in IFY Mathematics (Business/ Engineering/ Science) semester one, topic C.	C1	Solve inequalities such as $\frac{1}{x-a} > \frac{x}{x-b}$ that will reduce to a cubic inequality. Equations involving a modulus will not be set.
	C2	Sketch curves of the form $y = \frac{f(x)}{g(x)}$ where $f(x)$ and $g(x)$ are linear or quadratic functions.

TOPIC D – SERIES		
CONTENT	SLOs	
⇒ Summation of simple finite series. ⇒ The method of differences. ⇒ This section builds on the series work covered in IFY Mathematics (Business/ Engineering/ Science) semester one, topic E	D1	Sum series whose n^{th} term is given by a cubic polynomial at most.
	D2	Sum series using the method of differences.
	D3	Sum series with variable limits.

TOPIC E – ROOTS AND COEFFICIENTS OF POLYNOMIALS		
CONTENT	SLOs	
⇒ Relations between the roots and coefficients of quadratic and cubic equations. ⇒ Evaluation of symmetric functions in 2 or 3 variables.	E1	Form a new quadratic or cubic equation whose roots are related to the roots of a given equation (roots notation α , β and γ). Students will be expected to apply a suitable factorised form of $\alpha^3 + \beta^3$.
	E2	Evaluate symmetric functions of the roots using the sum and product and product two at a time of the roots.

TOPIC F – MECHANICS		
CONTENT	SLOs	
⇒ Kinematics: equations of motion for constant acceleration. ⇒ Newton's Laws. ⇒ Application of kinematics and Newton's Laws to connected particles.	F1	Identify displacement, velocity, time and acceleration.
	F2	Use the equations of motion for constant acceleration.
	F3	Appreciate that gravitational acceleration is constant and is independent of mass.
	F4	Understand Newton's Laws of Motion. Solve problems involving force, mass and acceleration.
	F5	Use and evaluate frictional forces.
	F6	Solve problems involving connected particles. (Pulleys will be smooth; and strings will be light and inextensible).

TOPIC G – HYPERBOLIC FUNCTIONS		
CONTENT	SLOs	
⇒ Definition of the six hyperbolic functions in terms of exponentials. ⇒ Graphs and properties of hyperbolic functions. ⇒ Inverse hyperbolic functions, their graphs, properties and logarithm equivalents. ⇒ This depends on exponential functions – covered in IFY Mathematics (Business/Engineering/ Science) semester one, topic D.	G1	Recognise $\sinh x$, $\cosh x$, $\tanh x$, $\operatorname{sech} x$, $\operatorname{cosech} x$ and $\coth x$ in terms of exponentials and connections between them e.g. $\tanh x = \frac{\sinh x}{\cosh x}$.
	G2	Derive simple identities such as $\cosh^2 x - \sinh^2 x \equiv 1$ and $\cosh^2 x + \sinh^2 x \equiv \cosh 2x$; and use them to solve equations such as $a \cosh x + b \sinh x = c$, $a \cosh^2 x + b \sinh x = c$.
	G3	Use and derive logarithmic equivalents of inverse hyperbolic functions e.g. $\sinh^{-1} x = \ln(x + \sqrt{x^2 + 1})$.
	G4	Know the differentiation and integration of $\sinh x$ and $\cosh x$.

TOPIC H – PARAMETRIC COORDINATES		
CONTENT	SLOs	
⇒ Parametric equations of curves and conversion between Cartesian and parametric forms. ⇒ Equations of tangents, normals and chords. Calculation of an area. ⇒ This depends on differentiation and integration – covered in IFY Mathematics (Business/Engineering/Science) semester one, topics G, H.	H1	Convert parametric equations into Cartesian form.
	H2	Find the area under a curve (which can include an ellipse) from the parametric equations Apply $\int f(y) \frac{dx}{dt} dt$ or $\int f(x) \frac{dy}{dt} dt$.
	H3	Find equations of tangents, normals and chords from the parametric equations.

TOPIC I – CONIC SECTIONS		
CONTENT	SLOs	
⇒ Cartesian and parametric equations for the parabola, ellipse and hyperbola. ⇒ The focus-directrix properties of the parabola, ellipse and hyperbola, including the eccentricity. ⇒ Equations of tangents, normals and chords to these curves. Simple loci problems. ⇒ This topic uses the geometry covered in IFY Mathematics (Business/ Engineering/ Science) semester one, topics A and C.	I1	Recognise: $y^2 = 4ax$, $x = at^2$, $y = 2at$ $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, $x = a \cos t$, $y = b \sin t$ $xy = c^2$, $x = ct$, $y = \frac{c}{t}$ $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, $x = a \sec t$, $y = b \tan t$ and $x = a \cosh t$, $y = b \sinh t$.
	I2	Intersection of a plane with a cone.
	I3	Calculate the eccentricity, find the coordinates of the foci and derive the equations of the directrices: for example, for the ellipse, apply $b^2 = a^2(1 - e^2)$, the foci are $(\pm ae, 0)$ and the equations of the directrices are $x = \pm \frac{a}{e}$.
	I4	Solve simple loci problems.

TOPIC J – MACLAURIN AND TAYLOR SERIES		
CONTENT	SLOs	
⇒ Third and higher order derivatives. ⇒ Derivation and use of Maclaurin and Taylor series. ⇒ This depends on differentiation – covered in IFY Mathematics (Business/ Engineering/ Science) semester one, topic G.	J1	Derive the series expansion of e^x , $\sin x$, $\cos x$, $\ln(1+x)$ and other simple functions.
	J2	Adding, subtracting and multiplying Maclaurin series to find more complicated series. Comparing coefficients to find Maclaurin series.
	J3	Derive the expansion e.g., of $\sin x$ in ascending powers of $(x - \pi)$ up to and including $(x - \pi)^3$.
	J4	Calculate numerical approximations to functions using these series.

TOPIC K – FURTHER MECHANICS		
CONTENT	SLOs	
⇒ Momentum, impulse and restitution involving up to a maximum of 3 collisions. ⇒ Kinetic and Potential Energy. ⇒ Work done and power. ⇒ Moments of a force.	K1	Find the impulse of a particle in one-dimension.
	K2	Apply the conservation of momentum principle.
	K3	Calculate impulse imparted in any impact.
	K4	Apply the coefficient of restitution in an impact.
	K5	Calculate Kinetic Energy and Potential Energy and solve problems where one type is converted either wholly or partly into another.
	K6	Find work done and calculate power output.
	K7	Find the moment of a force and understand the concept of clockwise and anti-clockwise moments.

TOPIC L – FURTHER COMPLEX NUMBER FUNCTIONS		
CONTENT	SLOs	
⇒ Euler's relation $ei\theta = \cos\theta + i\sin\theta$. ⇒ Relations between trigonometric functions and hyperbolic functions. ⇒ De Moivre's theorem. ⇒ Integration methods using trigonometric identities.	L1	Express a complex number in the form $re^{i\theta}$.
	L2	Use $\cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$ and $\sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$.
	L3	Understand the connection between trigonometric and hyperbolic functions e.g. calculating $\cos(x + iy)$ and $\cosh(x + iy)$ in the form $a + ib$.
	L4	Use De Moivre's theorem to find $\cos m\theta$ and $\sin m\theta$ in terms of powers of $\sin\theta$ and $\cos\theta$ and also powers of $\sin\theta$ and $\cos\theta$ in terms of multiple angles.
	L5	Integration of powers of $\sin\theta$ and $\cos\theta$.

TOPIC M – FURTHER DIFFERENTIATION AND INTEGRATION		
CONTENT	SLOs	
⇒ Differentiation and integration of hyperbolic functions; differentiation and integration of inverse trigonometric and inverse hyperbolic functions. ⇒ Integration using hyperbolic and trigonometric substitutions. ⇒ The calculation of arc length and the area of a surface of revolution.	M1	Differentiate expressions such as $\tanh 5x$, $x \cosh^2 x$, $\frac{\sinh 3x}{\sqrt{x+1}}$.
	M2	Differentiate inverse trigonometric and hyperbolic functions.
	M3	Calculate e.g. $\int \cosh^{-1} x \, dx$, $\int \tan^{-1} x \, dx$.
	M4	Integrate e.g. $\sqrt{x^2 + a^2}$, $\frac{1}{\sqrt{x^2 + a^2}}$, $\frac{1}{\sqrt{x^2 - a^2}}$ including those requiring hyperbolic and trigonometric substitutions.
	M5	Find the arc length and the area of a surface of revolution where the curve is given in Cartesian or parametric form.

TOPIC N – VECTORS		
CONTENT	SLOs	
⇒ The vector product $\mathbf{a} \times \mathbf{b}$ and the scalar triple product $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$. ⇒ The use of vectors in problems involving points, lines and planes. ⇒ The equation of a line in the form $(\mathbf{r} - \mathbf{a}) \times \mathbf{b} = \mathbf{0}$. ⇒ The equation of a plane in the forms $\mathbf{r} \cdot \mathbf{n} = p$, $\mathbf{r} = \mathbf{a} + s\mathbf{b} + t\mathbf{c}$.	N1	Interpret $ \mathbf{a} \times \mathbf{b} $ as an area and $ \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) $ as a volume.
	N2	Use equivalent Cartesian forms.
	N3	Solve problems including distances, intersections and angles between points, lines and planes (omit distance between skew lines).

TOPIC O – CALCULUS AND VECTORS		
CONTENT	SLOs	
⇒ Using calculus to find displacement, speed and acceleration in one-dimension using expressions involving time. ⇒ Using calculus to find displacement, velocity and acceleration using the $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ system of vectors. ⇒ Express forces using the $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ system of vectors.	O1	Evaluate displacement, velocity and acceleration from one-dimensional equations involving time.
	O2	Use the $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ system of vectors to solve problems involving displacement, velocity and acceleration.
	O3	Solve problems involving forces using the $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ system of vectors.

TOPIC P – DIFFERENTIAL EQUATIONS		
CONTENT	SLOs	
⇒ First Order differential equations: Use of the integrating factor in $y' + Pxy = Qx$. ⇒ The linear second order differential equation $ay'' + by' + cy = f(x)$ where a, b and c are real constants and the particular integral can be found by inspection or trial.	P1	Form differential equations and obtain particular solutions.
	P2	Use integrating factor $e^{\int Pdx}$.
	P3	Know the auxiliary equation may have real, distinct, equal or complex roots.
	P4	Know $f(x)$ will have one of the forms ke^{px} , a polynomial of degree at most two or $m\cos\omega x + n\sin\omega x$.
	P5	Be familiar with the terms 'complementary function' and 'particular integral'.
	P6	Solve equations such as $\frac{d^2y}{dx^2} + 6y = \cos 4x$.

TOPIC Q – PROOF BY INDUCTION		
CONTENT	SLOs	
⇒ Use the method of induction to prove a general result. (Questions involving matrix products will not be set).	Q1	Apply induction to prove the summation of a series.
	Q2	Apply induction to prove that an expression is divisible by a specified integer.
	Q3	Apply induction to prove the general term of a recurrence relation.

APPENDIX A - EXEMPLAR TEACHING PLAN

Please note that the table below allocates a minimum of 4 hours contact time per week. Study Centres may need to schedule additional time depending on the ability of each cohort.

Week	Hours	Topic	
1	1	A - Complex Numbers	Introduction to $\sqrt{-1}$. Solving quadratics with complex roots.
	1	A - Complex Numbers	Real and imaginary parts. Arithmetic with complex numbers (add, subtract, multiply, divide).
	1	A - Complex Numbers	Argand diagram, plotting $z = x + iy$. Modulus and argument. Extra worksheets on modulus argument form may be useful.
	1	A - Complex Numbers	The form $z = r(\cos \theta + i \sin \theta)$ where $r = z $, $\theta = \arg z$.
2	1	A - Complex Numbers	Fundamental Theorem of Algebra. z is a root implies z^* is a root if the coefficients of a polynomial are real. Coefficients will always be real.
	1	A - Complex Numbers	Solving cubics and quartics with complex roots by finding linear and quadratic factors (using the above). Cubics will be similar to cubics in maths. Quartics will need more practice.
	1	A - Complex Numbers	Restrictions on $ z $ and $\arg z$ and the loci that arise from these restrictions. A handout on these may save time.
	1	A - Complex Numbers	Sketching loci and regions on an Argand diagram.
3	1	B - Matrices	Introduction. Definition and order of a matrix.
	1	B - Matrices	Addition, subtraction, multiplication and the conditions for them. The matrices I and Z .
	1	B - Matrices	Square matrices: finding the determinant of a 2x2 and a 3x3 matrix and finding the inverse of a 2x2 matrix. Transposes of matrices.

	1	B - Matrices	Definitions of eigenvalues and eigenvectors.
4	1	B - Matrices	Characteristic equation, eigenvalues and eigenvectors of 2x2 matrices.
	1	B - Matrices	Characteristic equation 3x3 matrix.
	1	B - Matrices	Eigenvalues and eigenvectors of 3x3 matrices. Note that the inverse of a 3x3 matrix will not be required.
	1	B - Matrices	Consolidation of all work on matrices.
5	1	C - Further Curve Sketching and Inequalities	Vertical, horizontal and oblique asymptotes of curves of the form $y = \frac{f(x)}{g(x)}$, $f(x)$, $g(x)$ at most quadratic.
	1	C - Further Curve Sketching and Inequalities	Intercepts with axes. Intercepts of curve with asymptotes. Use of computer software or graphical calculator. Students could practise these using software as directed study.
	1	C - Further Curve Sketching and Inequalities	Sketching these curves.
	1	C - Further Curve Sketching and Inequalities	Solving cubic inequalities (and other inequalities that reduce to them). Solving inequalities using graphs and algebraic methods. Again, software can help with this. Equations involving a modulus will not be set.
6	1	D - Series	Use of $\sum f(r)$ notation. Use of standard formulae for $\sum r$, $\sum r^2$ and $\sum r^3$ and combinations of them.
	1	D - Series	Summations whose limits can be simple functions of n .
	1	D - Series	Method of differences for summation.
	1	E - Roots and Coefficients of Polynomials	The basic symmetric functions for 2 and 3 variables. Expressing more complicated symmetric functions in terms of these. Obtaining equations of quadratics or cubics with roots related to given equations.

7	1	F - Mechanics	Identification of displacement, velocity and acceleration. Connections between these quantities in relation to time.
	1	F - Mechanics	Introduction of equations of motion where the acceleration is constant.
	1	F - Mechanics	Further development of the equations of motion. Appreciation of the constant nature of gravitational acceleration.
	1	F - Mechanics	Practice at using the equations of motion.
8	1	F - Mechanics	Introduction to Newton's Laws of Motion. Develop the idea of a force. As for last week, practising examples will be time well spent in independent and directed study.
	1	F - Mechanics	Use simple examples which involve calculations of force, mass and acceleration.
	1	F - Mechanics	Extend the idea to the kinematics learnt the previous week.
	1	F - Mechanics	Introduction of friction. Using and understanding the coefficient of friction.
9	1	F - Mechanics	Extending forces and motion to slopes. Candidates should be aware of the definition of a particle.
	1	F - Mechanics	Solving problems involving connected particles: examples could be masses connected to each other over a pulley. Strings will be light and inextensible.
	1	F – Mechanics	Extending the idea of connected particles to slopes. Pulleys will always be smooth.
	1	F – Mechanics	Solving more complex problems with connected particles e.g., where the string breaks during a period of motion.
10	1	G - Hyperbolic Functions	Exponential definitions. Graphs. All graphs could be given as handouts.
	1	G - Hyperbolic Functions	Identities and their proofs.

	1	G - Hyperbolic Functions	Simple equations involving hyperbolic functions for example, $a \sinh x + b \sinh x = c$, $a \cosh^2 x + \sinh x = c$.
	1	G - Hyperbolic Functions	The inverse functions and their graphs. Logarithmic forms and their proofs.
11	1	H - Parametric Equations	Examples of using $x = f(t), y = g(t)$ to specify a curve. $\frac{dy}{dx} = \frac{g'(t)}{f'(t)}$ Use of software is recommended.
	1	H - Parametric Equations	Tangents, normals and chords. Trigonometric and hyperbolic examples. May need to cover the derivatives for $\sinh x$ and $\cosh x$ if not done earlier.
	1	H - Parametric Equations	Finding areas by $\int f(y) \frac{dx}{dt} dt$ or $\int f(x) \frac{dy}{dt} dt$. Integration at this stage confined to work covered in IFY Maths (Bus/Eng/Sci) topic H.
	1	H - Parametric Equations	Reverting to Cartesian form by eliminating t .
12	1	I - Conic Sections	Introduction by using the focus directrix properties. Review of parametric coordinates and general coordinate geometry from IFY Maths (distance between two points, parallel lines, etc) may be helpful.
	1	I - Conic Sections	Intersection of a plane with a cone.
	1	I - Conic Sections	Parabola $y^2 = 4ax$. Focus and directrix locations. Use of $x = at^2$ and $y = 2at$. Tangents, normals and chords in parametric form.
	1	I - Conic Sections	Introduction to simple loci problems.
13	1	I - Conic Sections	Ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. Focus and directrix locations. Use of $x = a \cos t$, $y = b \sin t$. Includes locus problems.
	1	I - Conic Sections	Tangents, normals and chords in parametric form. Area under ellipse using parametric form.

	1	I - Conic Sections	Rectangular Hyperbola $xy = c^2$. Focus and directrix locations. Use of $x = ct$, $y = \frac{c}{t}$. Hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ Focus and directrix locations. Use of $x = a \sec t$, $y = b \tan t$ and $x = a \cosh t$, $y = b \sinh t$.
	1	I - Conic Sections	Tangents, normals and chords in parametric form.
14	1	J - Maclaurin and Taylor Series	Derivation of a Maclaurin series by differentiation. In the End of Semester 1 test, any differentiation will be confined to what is expected in IFY Mathematics (Bus/Eng/Sci) semester one topic G. Candidates will be expected to apply further differentiation techniques (e.g. Chain Rule) in the final examination.
	1	J - Maclaurin and Taylor Series	Derivation of a Taylor series by differentiation.
	1	J - Maclaurin and Taylor Series	Adding, subtracting and multiplying Maclaurin series to find more complicated series. Comparing coefficients to find Maclaurin series (e.g. $1 = \cos x (a_0 + a_1x + a_2x^2 + \dots)$ to find $\sec x$).
	1	J - Maclaurin and Taylor Series	Taylor expansion about a point $x = a$ by use of the formula. $\Rightarrow$ Approximation of function values by using Maclaurin and Taylor series. $\Rightarrow$ Approximation of integrals by use of Maclaurin series. Technology can be used to compare graphs of $f(x)$ and its Maclaurin/Taylor series.
15	End of Semester 1 Test		
16	1	K - Further Mechanics	Momentum in one dimension. Conservation of momentum. Students will need to be aware that impulse is defined as change in momentum and the product of force and time.
	1	K - Further Mechanics	Solving problems involving collisions and applying the conservation of momentum principle.
	1	K - Further Mechanics	Finding the impulse imparted to a body.
	1	K - Further Mechanics	Solving problems which involve momentum, impulse and force.

17	1	K - Further Mechanics	Use of coefficient of restitution and its application in collisions. Questions which involve up to three collisions could be set.
	1	K - Further Mechanics	Further development of coefficient of restitution.
	1	K - Further Mechanics	Introduction to Kinetic and Potential Energy. Students should be aware of other types of energy (such as elastic) but questions involving these will not be set.
	1	K - Further Mechanics	Solving problems involving Kinetic and Potential Energy and extending this to work and energy. Students should be aware that work done is change in energy and the product of force and distance moved in the direction of the force.
18	1	K - Further Mechanics	Extending the previous session to power output. Time can be well spent in directed and independent study.
	1	K - Further Mechanics	Consolidation of work, energy and power. Students will be expected to know that power output is the rate of doing work and the product of force and velocity.
	1	K - Further Mechanics	Introduction to moments of a force. Solving problems involving balancing beams.
	1	K - Further Mechanics	Extending previous session to more complex problems. Examples could be finding the tension in wires in a vertical plane holding up awning.
19	1	L - Further Complex Numbers	Recap of modulus and argument. De Moivre's theorem in the form $(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$ (n any integer). Revision sheet if necessary or just use 2 hours for de Moivre.
	1	L - Further Complex Numbers	Expressing multiple angles of sine and cosine in terms of powers of sine and cosine.

	1	L - Further Complex Numbers	Expressing powers of sine and cosine in terms of multiple angles of sine and cosine.
	1	L - Further Complex Numbers	Integration of powers of sine and cosine. Revision of integration of multiple angles if necessary.
20	1	L - Further Complex Numbers	The p^{th} roots of unity in trigonometric form. Worksheet on plotting these on Argand diagram.
	1	L - Further Complex Numbers	Solving $z^p = a + ib$ where a and b are real. Modulus and argument again.
	1	L - Further Complex Numbers	Euler's relation: p^{th} roots of unity in exponential form.
	1	L - Further Complex Numbers	Derivation of the relationship between trigonometric and hyperbolic functions. Calculating $\cos(x + iy)$ and $\cosh(x + iy)$ in the form $a + ib$ for examples.
21	1	M - Further Differentiation and Integration	Derivatives of the basic hyperbolic functions and their inverses. Review of the differentiation methods in the IFY Maths module.
	1	M - Further Differentiation and Integration	Tangents, normals, maxima and minima.
	1	M - Further Differentiation and Integration	Integrals of the basic hyperbolic functions and their inverses. Review integration by parts. Integrals of the form $\int \frac{dx}{\sqrt{ax^2+bx+c}}$, $\int \cosh^{-1}x \, dx$.
	1	M - Further Differentiation and Integration	Review of Maths integration methods. Integration of inverse trigonometric functions. Integrals such as $\int x \tan^{-1}x \, dx$.
22	1	M - Further Differentiation and Integration	Integration by substitution. Substitution covered in Maths. Integrals such as $\int \sqrt{a^2 + x^2} \, dx$ and $\int \frac{x^2}{\sqrt{x^2+a^2}} \, dx$.
	1	M - Further Differentiation and Integration	Using hyperbolic and trigonometric substitution.

	1	M - Further Differentiation and Integration	Evaluation of area of a surface of revolution and arc length of curves in Cartesian form by use of the formulae. Worksheets for self-study may be used since all integration techniques will have been covered.
	1	M - Further Differentiation and Integration	Use of Parametric form.
23	1	N - Vectors	Recap or introduce the relevant vector topics in maths, dot product, equation of a line.
	1	N - Vectors	Vector cross product in $a b \sin\theta\hat{n}$ form and determinant form, and its use in calculating areas of triangles. Handout on 3x3 determinants could be used.
	1	N - Vectors	Scalar triple product $a.(b \times c)$ and its determinant form. Use in calculating volumes of tetrahedrons and parallelepipeds.
	1	N - Vectors	Consolidation of previous sessions.
24	1	N - Vectors	The plane in 3 dimensions. Normal to a plane.
	1	N - Vectors	The Cartesian, scalar product and vector equations for a plane and converting between them.
	1	N - Vectors	Vector geometry. Distances: point from plane, point from line, between parallel planes and lines. A list of formulae and their derivations could be given as a handout. Problems involving two or more of these also. Distances between skew lines are excluded.
	1	N - Vectors	Intersections: line and plane, two planes. Angles: line and plane, two planes.
25	1	O – Calculus and Vectors	Evaluation of displacement, velocity and acceleration (which is not constant) in one dimension using calculus. The equations could include trigonometric and exponential expressions of time.
	1	O – Calculus and Vectors	Extending the above using the i, j and k system of vectors.

			Time spent doing examples as part of directed and independent study will be useful.
	1	O – Calculus and Vectors	Solving more complex problems using vectors.
	1	O – Calculus and Vectors	Expressing forces in vector form and applying Newton's Second Law.
26	1	O – Calculus and Vectors	Developing forces further and connecting with kinematics with vectors.
	1	O – Calculus and Vectors	Consolidation of previous five sessions.
	1	P - Differential Equations	The first order equation $y' + P(x)y = Q(x)$. Multiplying by integrating factor $e^{\int P(x)dx}$.
	1	P - Differential Equations	General and particular solutions.
27	1	P - Differential Equations	The linear second order equation $ay'' + by' + cy = f(x)$. Auxiliary equation and the nature of its roots determining the complementary function. Roots may be real (distinct or equal) or complex.
	1	P - Differential Equations	Further development of the complementary function.
	1	P - Differential Equations	The particular integral depending on $f(x)$. General solution = particular integral + complementary function. Particular solution. Handouts for self-study with lots of different combinations of $f(x)$ and the CF recommended.
	1	P - Differential Equations	Further development of particular integrals.
	1	P - Differential Equations	Consolidation on second order equations.
28	1	Q – Proof by Induction	Prove the summation of a series e.g., Use induction to prove that $\sum_{r=1}^n \frac{1}{r(r+1)} = \frac{n}{n+1}$
	1	Q – Proof by Induction	Use induction to prove that an expression is divisible by a specified integer. e.g., Use induction to prove that $9^n - 1$ is a multiple of 8 for all positive integral values of n .
	1	Q – Proof by Induction	

	1	Q – Proof by Induction	Use induction to prove the general term of a recurrence relation. e.g. Given that $u_{n+1} = 2u_n + 1$ and that $u_1 = 1$, prove by induction that $u_n = 2^n - 1$ Questions involving matrix products will not be set.
29	4	Revision	
30	Exam Week		

APPENDIX B – NOTATION LIST

See also the relevant notation list for Maths Business, Maths Science or Maths Engineering.

The notation below may be used in IFY Further Maths assessments without explanation. If forms of notation other than those listed are used in an assessment they will be defined within the question.

$\sec, \csc, \cot$	further trigonometric functions
$\sinh, \cosh, \tanh$	the hyperbolic functions
$\sinh^{-1}, \cosh^{-1}, \tanh^{-1}$	the inverse hyperbolic functions
i	square root of -1
$a + bi$	cartesian form of a complex number
$r(\cos\theta + i\sin\theta)$	modulus argument form of a complex number
z, w	a complex number
z^*	the complex conjugate of z
$\mathbf{M}$	a matrix $\mathbf{M}$
$\mathbf{M}^{-1}$	the inverse of the matrix $\mathbf{M}$
$\mathbf{M}^T$	the transpose of the matrix $\mathbf{M}$
$\mathbf{I}$	the identity matrix
$\mathbf{Z}$	the null matrix
α, β, γ	the roots of a quadratic/cubic equation
g	acceleration due to gravity
μ	coefficient of friction
$\mathbf{i}, \mathbf{j}, \mathbf{k}$	unit vectors in the directions of the cartesian coordinate axes

APPENDIX C - FORMULAE BOOKLET

Refer to separate Formulae Booklet.

It is the centre's responsibility to print "clean" (new) copies of the Formula Booklet for the End of Semester 1 Test and Final Examination.

Syllabus Update History: NCUK International Foundation Year Further Maths

Release Date	Version No.	Summary of Main Changes	Author
June 2024	4.4	Updated for 2024/25	Product Team
June 2023	4.3	Updated for 2023/24	Product Team
May 2022	4.2	Updated for 2022/23	Product Team
Sept 2021	4.1	Link to electronic version of textbook updated after change by publisher	Product Team
June 2020	4.0	Full Review	Product Team

This syllabus is valid for the 2024-25 academic year only.

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